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ERL-BCP: A balance-controlled plasticity network with energy-regulated liquid timescales

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ABSTRACT

Biologically plausible credit assignment requires local learning signals generated from circuit variables while maintaining stable continuous-time dynamics. However, continuous-time local learning networks often face coupled challenges arising from fixed membrane timescales, load-insensitive current regulation, and potentially unbounded feedback signals. To address these challenges, we introduce a balance-controlled plasticity network with energy-regulated liquid timescales (ERL-BCP), a continuous-time local learning framework that further incorporates bounded inhibitory feedback. In each assembly, the squared magnitudes of excitatory, input-associated inhibitory, and balance-sensitive inhibitory currents define a local current energy. This energy increases membrane decay under high current load and shortens the effective timescale, while a smoothly saturated controller limits task-driven inhibitory feedback. Synaptic updates remain local and are driven by filtered E/I balance deviations. Under the stated assumptions, the energy-dependent dynamics admit a uniform bound on the excitatory membrane state. Across five static, event-based, and sequential benchmarks, ERL-BCP achieves the highest mean best test accuracy among the evaluated local methods in every tested configuration and maintains stable performance across the evaluated network widths and depths. Controlled dynamical probes further show that increasing current load shortens the effective liquid timescale and that shorter timescales accelerate membrane-state establishment and perturbation recovery. Controlled variants show that replacing the balance-derived signal causes the largest deterioration in predictive performance and internal regulation, fixing the timescale reduces accuracy and weakens balance regulation, and removing the feedback bound markedly increases current energy and balance residual with little change in accuracy. At matched parameter states and probe batches, the ERL-BCP update is more closely aligned with supervised descent than a generic local update in all 155 paired evaluations per dataset.

1. Introduction

Biologically plausible learning addresses a central question in neural computation: how can neural systems derive effective learning signals from variables that are available to neurons, synapses, and local circuits themselves? This question is relevant not only to biological realism, but also to the design of learning systems that avoid globally coordinated error transport (Ororbia et al., 2024). A broad family of local learning approaches has pursued this goal by replacing explicitly backpropagated gradients with locally defined prediction errors (Millidge et al., 2022), equilibrium differences (Scellier et al., 2023), target discrepancies (Bengio, 2020), perturbation responses (Ororbia et al., 2023), or correlation-based signals (Laborieux et al., 2021). These methods differ in their objectives and circuit interpretations, but they share a common premise: effective plasticity can be driven by local variables when those variables carry task-relevant information and arise from well-defined network dynamics.

Within this family, excitation-inhibition (E/I) balance offers a particularly concrete circuit-level substrate for local learning. Balance-based approaches ground instructive signals in the relation between excitatory drive and inhibitory compensation, giving learning a direct microcircuit interpretation (Barzon et al., 2025). This is attractive because E/I balance is closely associated with stability, responsiveness, selectivity, and controlled population dynamics (Liang et al., 2025; Tan et al., 2004; Wehr & Zador, 2003; Xue et al., 2014). However, in much of the E/I balance research, balance has primarily been treated as a regulatory target, and its role in conveying task-relevant signals remains insufficiently understood (Denève & Machens, 2016; Hennequin et al., 2017;

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Meissner-Bernard et al., 2025). Recent work on balance-controlled plasticity (BCP) makes this connection explicit by showing that local deviations from precise E/I balance can encode neuron-specific error signals for synaptic plasticity (Rossbroich & Zenke, 2025). In BCP, task information is delivered through top-down feedback to inhibitory pathways, producing balance perturbations that determine the sign and magnitude of local synaptic change. The learning signal is therefore expressed as a deviation of the local circuit from its balanced state.

Despite this advance, the standard BCP formulation retains a fixed membrane timescale within each assembly (Rossbroich & Zenke, 2025).

Recent dynamical modeling studies have highlighted the importance of multiscale and state-dependent neural regulation. Fast–slow astrocytic dynamics have been shown to coordinate neural processes across memory timescales (Yin et al., 2026), while cortico-cerebellar dynamics couple decision processes with error-driven temporal adaptation (Shen et al., 2026), and hippocampal-basal ganglia interactions support dynamic shifts between navigation strategies during learning (Wei et al., 2025).

The balance deviation that drives plasticity is generated by membrane states, firing rates, excitatory currents, inhibitory currents, and feedback pathways that evolve continuously in time (Rossbroich & Zenke, 2025). Yet, when these dynamics use a fixed membrane timescale, the assembly responds with the same temporal mode under weak input, strong input, transient perturbation, persistent feedback error, and different stages of learning (Hasani et al., February 2–9, 2021). This limits the adaptability of the learning substrate: the plasticity signal remains local in form, but the dynamical process that generates it does not adjust to the assembly's current operating condition. For balance-based learning, this limitation becomes important when input statistics, task demands, or internal current regimes vary over time.

A second limitation is that E/I balance alone does not constrain either the absolute current magnitude within the network or the amplitude of task-driven feedback. Because balance is a relative condition, excitation and inhibition may both become large while their residual mismatch remains small (Rubin et al., 2017). The finite operating ranges of membrane potentials, firing rates, synaptic currents, and metabolic resources provide a biological motivation for avoiding such high-current states (Attwell & Laughlin, 2001; Howarth et al., 2012). Computationally, large opposing currents can increase perturbation sensitivity, degrade numerical conditioning, and place the network in an unnecessarily high-gain regime (Hennequin et al., 2014). Persistent output errors can further inject sustained control currents through the inhibitory feedback pathway unless the feedback amplitude is explicitly bounded. These limitations motivate two complementary extensions to BCP: local membrane timescales should adapt to the current load within the network, and task feedback should perturb inhibitory balance within a controlled amplitude range.

We address these limitations by introducing a balance-controlled plasticity network equipped with energy-regulated liquid timescales (ERL-BCP). This framework preserves the central BCP principle that synaptic plasticity is driven by local deviations from E/I balance and introduces two extensions to the underlying dynamics. First, the temporal regime of each local assembly becomes state dependent. Each assembly decomposes its drive into excitatory, input-associated inhibitory, and balance-sensitive inhibitory current components. The sum of their squared magnitudes defines an assembly-level current energy that summarizes the local electrical operating condition. This energy modulates the effective decay rate and liquid timescale, allowing the assembly to adjust its temporal responsiveness to its current load. Second, ERL-BCP introduces bounded inhibitory feedback. Output-level task error is transformed into a smoothly saturated control signal before entering the balance-sensitive inhibitory population. The energy-regulated timescale increases membrane contraction under high current load, while feedback saturation limits task-driven current injection under persistent error. Together, these mechanisms regulate the dynamics through which local balance deviations are generated. This work makes three main contributions.

- **Energy-regulated Liquid Timescales.** We introduce an assembly-level current-load mechanism that continuously adjusts the effective membrane timescale of each local E/I circuit according to its instantaneous operating state. This state-dependent temporal adaptation enriches within-assembly dynamics.
- **Bounded Inhibitory Feedback.** We model task supervision as a smooth, saturated disturbance current in the inhibitory pathway. This form of feedback not only shapes local equilibrium deviations but also limits the sustained current injection caused by output errors and the excessive growth of the closed-loop current.
- **Empirical Validation.** We evaluate ERL-BCP through task-performance comparisons, direct dynamical characterization of the energy-regulated liquid timescale, controlled mechanism variants, and update alignment analysis. ERL-BCP achieves strong and consistent test performance, maintains lower balance residual and current energy, and produces local updates that align more closely with supervised descent than a matched generic local error.

2. Related work

2.1. Local learning and biologically plausible credit assignment

Biologically plausible learning seeks to solve credit assignment without relying on the exact global gradients required by backpropagation (Ororbia et al., 2024). This objective has motivated a broad class of local learning approaches in which synaptic updates are driven by variables available to neurons, synapses, or local circuit modules. Predictive coding uses hierarchical prediction errors as local learning signals, with higher-level activity generating predictions and lower-level mismatches driving inference and parameter updates (Millidge et al., 2022; Rao & Ballard, 1999; Song et al., 2020). Equilibrium propagation obtains local updates from the difference between neural activities in a free equilibrium state and a weakly perturbed equilibrium state (Laborieux et al., 2021; Scellier & Bengio, 2017; Scellier et al., 2023). Correlative information maximization provides another recent route, formulating supervised learning through correlative mutual information between adjacent-layer activities and deriving asymmetric prediction pathways with local activity-dependent plasticity (Bozkurt et al., 2023; Erdogan, 2022). These methods establish that task-relevant credit assignment can be approximated through locally available variables. Yet their main focus is the construction or optimization of local learning signals. The dynamical regime in which such signals are produced, particularly when they arise from continuously evolving membrane states and synaptic currents, remains comparatively underexplored.

2.2. E/I balance learning

E/I balance theory states that, under appropriate coupling between excitatory and inhibitory neurons, recurrent neural circuits can enter a regime in which strong excitation and inhibition approximately cancel, leaving residual fluctuations to drive irregular, noise-like spiking activity (Renart, 2010; Van Vreeswijk & Sompolinsky, 1996). This balanced operating regime has been linked not only to the prevention of unstable activity, but also to efficient information processing (Barzon et al., 2025), complex computation (Shew & Yang, 2009), and heightened sensitivity to sensory inputs. Because balanced dynamics emerge from the interaction between strong recurrent excitation, compensatory inhibition, and population-level connectivity, they provide a powerful but analytically challenging framework for understanding local circuit computation.

More recent work has moved E/I balance beyond its traditional role as a stabilizing constraint or coding regime, treating deviations from

balance as instructive signals for synaptic plasticity. BCP (Rossbroich & Zenke, 2025) encodes neuron-specific error signals through deviations from local E/I balance, while feedback to inhibitory pathways shapes supervised learning during these deviations. This gives local error-driven learning a concrete microcircuit interpretation, with synaptic change determined by how a local E/I assembly is displaced from its balanced state. Yet existing balance-based formulations mainly specify how balance deviations are produced and used for plasticity. They leave comparatively open how the membrane dynamics that generate those deviations should adapt when the same assembly operates under different current loads, input conditions, or learning stages.

2.3. Continuous-time neural network models

Continuous-time neural network models constitute a class of computational frameworks used to study how coupled neural states evolve under sequential inputs. Continuous-time models can approximate rich dynamical systems and support state-dependent integration, transient memory, and adaptive responses to perturbations (Chen et al., 2018; Funahashi & Nakamura, 1993). Ordinary differential equation models further formalize learning in continuous-depth or continuous-time systems, providing flexible tools for modeling dynamical transformations and temporal computation (Chen et al., 2018; Wang , 2026b). These models are relevant to biologically inspired learning because membrane potentials, firing rates, and synaptic currents are inherently time-evolving quantities rather than purely discrete computational states.

More recent liquid time-constant networks and neural circuit policies extend this perspective by allowing effective temporal behavior to vary with the state and input of the system (Chahine et al., 2023; Hasani et al., February 2–9, 2021). Such networks show that timescale can improve temporal processing, robustness, and control in dynamic environments. However, liquid-dynamics networks are typically developed as general continuous-time architectures for sequence modeling, control, or interpretable recurrent computation. Their timescale modulation is usually not tied to the current decomposition of a local E/I assembly, and it is not formulated around balance deviations as plasticity-driving signals. Conversely, balance-controlled learning provides a biologically grounded local error substrate, but commonly assumes a fixed membrane timescale for the dynamics that generate that substrate. This leaves a conceptual gap between adaptive continuous-time dynamics and E/I-balance-based local plasticity: how the temporal regime of a balance-generating circuit should be regulated by its own operating condition remains largely unexplored.

3. Methods

3.1. ERL-BCP framework

ERL-BCP is formulated as a continuous-time rate-based artificial neural network with biologically inspired E/I organization. At the unit level, each dynamic neuron is described by a continuous-time state and a rate output. These neurons are organized into local E/I assemblies that serve as the basic computational components of the network. At the network level, the coupled E/I assemblies, energy-regulated liquid timescales, bounded inhibitory feedback, and balance-controlled local plasticity together constitute the complete ERL-BCP framework.

Fig. 1 summarizes the ERL-BCP architecture, which consists of three coupled components. First, each balance-controlled plasticity assembly decomposes the excitatory membrane drive into excitation-related, input-associated inhibitory, and balance-sensitive inhibitory current components. These currents jointly define the local net drive, the balance deviation used for plasticity, and the assembly-level current energy used for timescale regulation. Second, the instantaneous current energy modulates the effective decay rate and liquid timescale, thereby regulating membrane-potential evolution under different current loads. Third, during the feedback phase, output-level task error is converted into a bounded inhibitory feedback signal and projected to the balance-sensitive inhibitory population. The updated membrane states generate firing rates that reshape recurrent currents and local balance conditions, while the resulting balance deviation drives synaptic plasticity through a local pre- and postsynaptic update rule. Thus, current decomposition specifies the circuit state, energy-regulated timescales control temporal responsiveness, bounded inhibitory feedback conveys task information, and the balance deviation serves as the local plasticity signal.

Let a denote a local assembly and i index a neuron within the population specified by the corresponding population label. The membrane-like states of the excitatory and balance-sensitive inhibitory populations are denoted by $u_i^{E,(a)}(t)$ and $u_i^{I_2,(a)}(t)$, respectively, with corresponding rate outputs

$$r_i^{E,(a)}(t) = \phi_E\left(u_i^{E,(a)}(t)\right), r_i^{I_2,(a)}(t) = \phi_I\left(u_i^{I_2,(a)}(t)\right). \tag{1}$$

Here, r and denotes the rate output of a dynamic neuron, while ϕ_E and ϕ_I denote the activation functions of the E and I_2 populations, respectively. The E and I_2 populations possess explicit continuous-time membrane states, whereas I_1 is implemented as an input-associated inhibitory current pathway without an independent neuronal state. Both dynamic populations use the Softplus activation in the reported experiments.

3.2. Balance-controlled plasticity assembly

The balance-controlled plasticity assembly component of ERL-BCP primarily provides neuron-specific signals for synaptic plasticity through local deviations from the E/I balance. This section defines current decomposition, balance deviations, and local plasticity rules, which constitute the circuit substrate for the energy-regulated liquid timescale and bounded inhibitory feedback introduced in Sections 3.3 and 3.4.

For each excitatory neuron i in assembly a, the local E/I circuit generates three current components that determine its membrane drive and local balance condition: an excitation-related current $I_i^{\mathrm{exc},(a)}(t)$, an input-associated inhibitory current $I_i^{\mathrm{inh1},(a)}(t)$, and a balance-sensitive inhibitory current $I_i^{\mathrm{inh2},(a)}(t)$. In general, $w_{ij}^{P\to Q,(a)}$ denotes the coupling from presynaptic source P to postsynaptic population Q within assembly a. The E and I_2 populations are represented as dynamic neuronal populations.

$$I_i^{\mathrm{exc},(a)}(t) = \sum_j w_{ij}^{X\to E,(a)} x_j(t) + \sum_k w_{ik}^{E\to E,(a)} r_k^{E,(a)}(t), \tag{2}$$

The I_1 pathway is input-associated, with $w^{X\to I_1,(a)}$ denoting its effective coupling coefficients. Its instantaneous contribution is computed directly from the external or preceding-layer input as

$$I_i^{\mathrm{inh1},(a)}(t) = \sum_j w_{ij}^{X\to I_1,(a)} x_j(t), \tag{3}$$

The balance-sensitive inhibitory current is generated by the dynamic I_2 population. Its membrane dynamics are defined as

$$\frac{du_i^{I_2,(a)}(t)}{dt} = -\frac{1}{\tau_I} u_i^{I_2,(a)}(t) + I_{I_2,i}^{\mathrm{exc},(a)}(t) - I_{I_2,i}^{\mathrm{inh1},(a)}(t) + I_i^{\mathrm{fb},(a)}(t), \tag{4}$$

$$I_i^{\mathrm{inh2},(a)}(t) = \sum_j w_{ij}^{I_2\to E,(a)} r_j^{I_2,(a)}(t). \tag{5}$$

Here, $\tau_I > 0$ is the fixed inhibitory membrane timescale and is not modulated by the assembly-level current energy. $I_{I_2,i}^{\mathrm{exc},(a)}(t)$ denotes the excitation-related drive received by the I_2 neuron, including feedforward input and recurrent $E \to I_2$ excitation, whereas $I_{I_2,i}^{\mathrm{inh1},(a)}(t)$ denotes its input-associated inhibitory contribution. The task-dependent current $I_i^{\mathrm{fb},(a)}(t)$ perturbs the I_2 membrane state through the bounded feedback mechanism described in Section 3.4. Accordingly, $I_i^{\mathrm{exc},(a)}(t)$, $I_i^{\mathrm{inh1},(a)}(t)$, and $I_i^{\mathrm{inh2},(a)}(t)$ constitute the three current components received by the

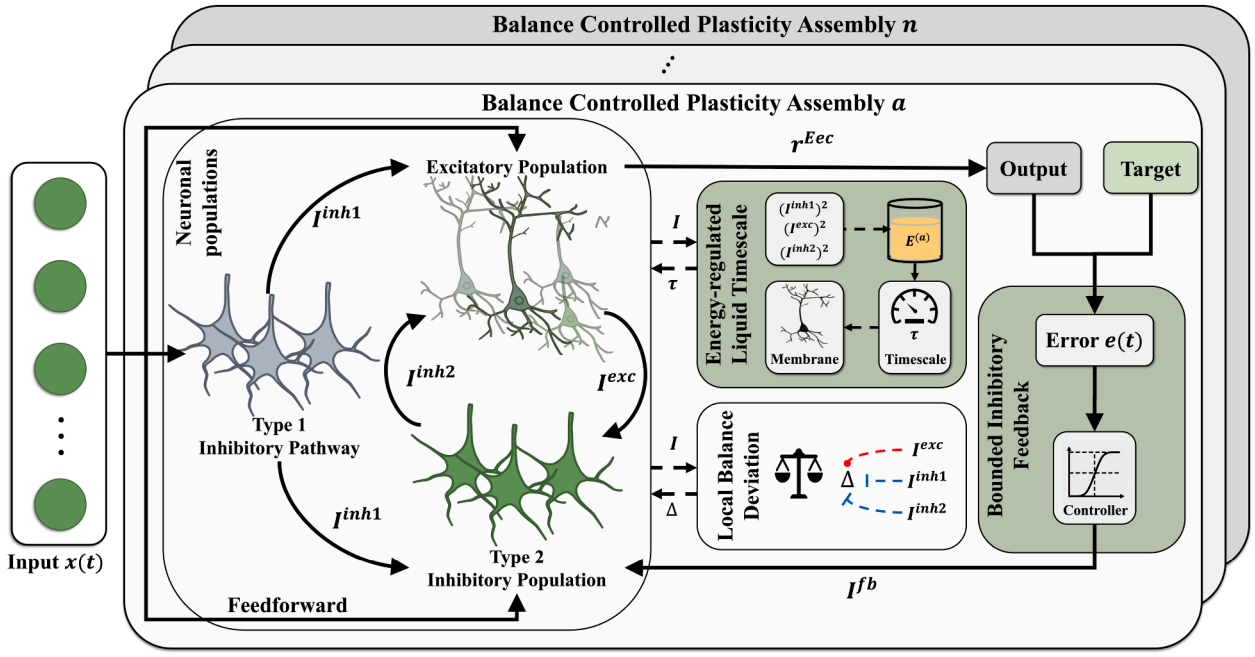


Fig. 1. Architecture and information flow of ERL-BCP. Each balance-controlled plasticity assembly contains an excitatory population, an input-associated inhibitory current pathway, and a balance-sensitive inhibitory population. The three current components received by the excitatory population define the local balance deviation used for synaptic plasticity and the assembly-level current energy used to regulate the liquid timescale. Output error is transformed by a bounded controller and projected to the balance-sensitive inhibitory population, allowing task information to perturb local E/I balance while limiting feedback-induced current injection.

excitatory population and are subsequently used to define its local balance condition and net membrane drive.

The recurrent E/I_2 pathways follow a Dale-like sign convention. $W^{E\to I_2,(a)}$ and $W^{I_2\to E,(a)}$ are represented by nonnegative magnitudes, while their excitatory and inhibitory effects are imposed by fixed pathway signs; these recurrent couplings remain fixed during ERL-BCP training.

To reduce transient mismatch caused by asynchronous current evolution, the excitation-related and input-associated inhibitory currents are low-pass filtered. For $I_i^{q,(a)}(t)$, where $q \in \{\text{exc}, \text{inh1}\}$, we define

$$\tau_q \frac{d\bar{I}_i^{q,(a)}(t)}{dt} = -\bar{I}_i^{q,(a)}(t) + I_i^{q,(a)}(t), \tag{6}$$

where $\tau_q > 0$ is the filtering timescale. All variables are represented as dimensionless computational quantities; the corresponding convention is stated explicitly in Appendix B.

The local balance deviation $\Delta_i^{(a)}(t)$, hereinafter referred to as the "balance-derived signal", is then defined as

$$\Delta_i^{(a)}(t) = \alpha\left(\bar{I}_i^{\text{exc},(a)}(t) - \bar{I}_i^{\text{inh1},(a)}(t)\right)_+ - I_i^{\text{inh2},(a)}(t), \tag{7}$$

where $(\cdot)_+$ denotes rectification and $\alpha > 0$ specifies the target proportional relation between excitation-related drive and balance-sensitive inhibitory compensation.

A positive $\Delta_i^{(a)}(t)$ indicates insufficient balance-sensitive inhibition relative to the filtered local drive, whereas a negative value indicates excessive compensatory inhibition. The signal is computed within the assembly and is not a direct copy of the output error. In ERL-BCP, assembly energy regulates membrane dynamics, bounded inhibitory feedback controls task-driven perturbation, and $\Delta_i^{(a)}(t)$ supplies the local signal for synaptic plasticity.

The feedforward excitatory synapses are updated by

$$\frac{dw_{ij}^{X\to E,(a)}(t)}{dt} = \eta \bar{x}_j(t)\Delta_i^{(a)}(t), \tag{8}$$

where $\eta > 0$ is the learning rate and $\bar{x}_j(t)$ is a filtered presynaptic trace. The update is local because it depends only on a presynaptic variable and a postsynaptic balance deviation computed within the same assembly.

3.3. Energy-regulated liquid timescale

In circuit-derived neural network models, equations defined from electrical components can be expressed as dimensionless dynamical systems, with corresponding energy functions defined under appropriate scaling (Ma, 2025; Wang et al., 2026a). ERL-BCP adopts a dimensionless computational formulation without constructing an equivalent physical circuit. Accordingly, the assembly-level quantity introduced below is defined from dimensionless current-like variables as a measure of local current load. A fixed membrane timescale imposes a common temporal regime across varying current conditions. Motivated by conductance-based neuronal dynamics, in which increased synaptic conductance shortens the effective membrane time constant according to $\tau_{\text{eff}} = C_m/g_{\text{tot}}$, and by evidence that neural timescales vary with ongoing behavioral demands (Manea et al., 2024; Sacha et al., 2025), ERL-BCP uses this load measure to regulate the effective decay rate and timescale.

The quadratic dependence is motivated by the I^2R form of resistive dissipation (Jin et al., 2019), while $E^{(a)}(t)$ is used here solely as a dimensionless measure of absolute current load. Squaring the pathway currents yields a nonnegative and sign-insensitive quantity that remains informative when large opposing excitatory and inhibitory currents produce only a small signed residual. Accordingly,

$$E^{(a)}(t) = \sum_{i=1}^{N_a}\sum_{\upsilon} R_\upsilon (I_i^{\upsilon,(a)}(t))^2, \upsilon \in \{\text{exc}, \text{inh1}, \text{inh2}\}. \tag{9}$$

Summing the three pathway contributions provides an assembly-level measure of absolute current load that complements the signed balance deviation used for plasticity. The coefficients R_υ specify the relative contribution of each current pathway. In all reported simulations, they are set to unity, $R_{\text{exc}} = R_{\text{inh1}} = R_{\text{inh2}} = 1$, which recovers the squared-current calculation used in the original implementation.

The energy-regulated decay rate is defined as

$$\lambda^{(a)}(t) = \frac{1}{\tau_{\text{base}}} + \mu E^{(a)}(t), \qquad \mu > 0, \tag{10}$$

where $\tau_{\text{base}} > 0$ is the baseline membrane timescale and μ controls the strength of energy regulation. The energy-regulated liquid timescale is

$$\tau^{(a)}(t) = \frac{1}{\lambda^{(a)}(t)} = \frac{\tau_{\text{base}}}{1 + \mu\tau_{\text{base}} E^{(a)}(t)} \tag{11}$$

Since $E^{(a)}(t) \geq 0$, the timescale satisfies

$$0 < \tau^{(a)}(t) \leq \tau_{\text{base}}. \tag{12}$$

The resulting net drive to the membrane state is

$$D_i^{(a)}(t) = I_i^{\text{exc},(a)}(t) - I_i^{\text{inh1},(a)}(t) - I_i^{\text{inh2},(a)}(t). \tag{13}$$

This decomposition defines the net membrane drive. The membrane-potential dynamics of excitatory neurons are then given by

$$\frac{du_i^{E,(a)}(t)}{dt} = -\lambda^{(a)}(t) u_i^{E,(a)}(t) + D_i^{(a)}(t) \tag{14}$$

$$= -\lambda^{(a)}(t)[u_i^{E,(a)}(t) - \tau^{(a)}(t) D_i^{(a)}(t)]. \tag{15}$$

Equivalently, $\lambda^{(a)}(t) = 1/\tau^{(a)}(t)$ determines the relaxation rate of the excitatory membrane state, while $u_{i,\star}^{(a)}(t) = \tau^{(a)}(t) D_i^{(a)}(t)$ defines its instantaneous target.

Hereafter, Eqs. (10)–(15), which jointly define the current-energy-dependent decay rate, effective timescale, and instantaneous target, are referred to as the ERL dynamics. The energy regulator therefore affects both temporal relaxation and the instantaneous drive-to-state gain. The dynamical effect of the regulated timescale is characterized directly in Section 4.2, while the separate contributions of adaptive temporal relaxation and Dynamic Gain are evaluated through controlled variants in Section 4.3.

This construction also contributes to stable operation. The same current components that form the net drive also enter the decay rate through a quadratic energy term. Consequently, high-current regimes increase the contraction of the membrane dynamics, limiting the accumulation of large drive magnitudes in the membrane state. Under bounded inputs and finite weights, this coupling yields a bounded instantaneous membrane target and a forward-invariant membrane interval. The detailed boundedness argument is provided in Appendix A.

3.4. Bounded inhibitory feedback

BCP expresses task supervision by perturbing local balance through top-down feedback. ERL-BCP retains this route while constraining the controller output before it reaches the balance-sensitive inhibitory population. The bounded interface limits sustained feedback drive that could otherwise dominate the local current decomposition and distort the balance-derived signal.

Let $e(t)$ denote the output-level task error. A leaky proportional-integral controller first forms an unconstrained control variable:

$$\hat{c}(t) = K_P e(t) + K_I z(t), \qquad \frac{dz(t)}{dt} = -\rho z(t) + e(t), \tag{16}$$

where K_P and K_I are proportional and integral gains, $z(t)$ is the leaky integral state, and $\rho > 0$ controls integral leakage. The controller is applied componentwise when the output error is vector valued. The unconstrained control is then passed through a smooth saturation:

$$c^{\text{sat}}(t) = L_c \tanh\left(\frac{\hat{c}(t)}{L_c}\right), \tag{17}$$

where $L_c > 0$ sets the maximum feedback amplitude.

The saturated control signal is projected to the balance-sensitive inhibitory channel of each assembly:

$$I_i^{\text{fb},(a)}(t) = \sum_m B_{im}^{(a)} c_m^{\text{sat}}(t), \tag{18}$$

where $B^{(a)}$ denotes the output to inhibitory feedback projection for assembly a. It is recomputed for each batch from the current network Jacobian evaluated at the open-loop terminal state. Specifically,

$$B^{(a)} = \left(J^{(a)} M_I^{(a)\top}\right)^\top, \qquad J^{(a)} = \frac{\partial y_{\text{out}}}{\partial h^{(a)}}, \tag{19}$$

where $h^{(a)}$ denotes the hidden representation associated with assembly a and $M_I^{(a)}$ is the corresponding inhibitory membership mapping. The resulting projection is normalized to unit Frobenius norm before applying Eq. (18). The normalized projection injects the bounded task signal into the I_2 population, thereby perturbing the local balance condition while leaving the synaptic update itself local.

The saturation level L_c controls the maximum amplitude of task-driven perturbation. For small errors, $c^{\text{sat}}(t)$ behaves approximately linearly and preserves the corrective role of feedback. For persistent or large errors, the saturation prevents unbounded control currents from entering the inhibitory pathway. ERL-BCP thereby retains feedback-guided balance perturbation while constraining the feedback interface during continuous-time learning.

4. Experiments

We evaluate ERL-BCP from four complementary perspectives. First, task-performance experiments compare ERL-BCP with representative local learning methods and backpropagation based references across static, event-based, and sequential benchmarks. Second, controlled dynamical probes directly characterize how current load regulates the effective liquid timescale and how the resulting timescale determines membrane-state response rates. Third, controlled mechanism variants isolate the contributions of the balance-derived signal, energy-regulated liquid timescale, Dynamic Gain, and bounded inhibitory feedback to predictive performance and closed-loop dynamics. Fourth, directional-alignment analysis examines whether the balance-derived local update is consistently better aligned with supervised descent than a matched generic local update.

4.1. Task performance

We assess ERL-BCP across static, event-based, and sequential inputs and across network configurations that vary in width and depth. Experiments use MNIST (Deng, 2012), Fashion-MNIST (F-MNIST) (Xiao et al., 2017), Neuromorphic-MNIST (N-MNIST) (Orchard et al., 2015), Permuted-MNIST (P-MNIST), and Sequential MNIST (S-MNIST) (Le et al., 2015) with hidden-layer configurations of 300×1, 128×2, and 128×3, where the two terms denote hidden width and number of hidden layers. ERL-BCP is compared with three representative local learning methods: BCP, which uses E/I balance deviations for local plasticity; PC-Nudge (Millidge et al., 2023), which constructs local prediction errors from nudged states; and CorInfoMax (Bozkurt et al., 2023), which optimizes layer-local correlative information objectives. Backpropagation-trained MLP and RNN provide nonlocal performance references. For N-MNIST, P-MNIST, and S-MNIST, the comparison is restricted to recurrent networks that explicitly process temporal or sequential inputs. Each method and network configuration is evaluated over 30 independent runs with different random seeds.

We evaluate each run using the best test accuracy attained during training,

$$\text{Acc}_{\text{best}} = \max_t \text{Acc}(t), \tag{20}$$

and the normalized area under the learning curve,

$$\text{AUC} = \frac{1}{K}\sum_{k=1}^{K} \text{Acc}(t_k), \tag{21}$$

where t_k denotes the kth of K equally spaced evaluation epochs. For equally spaced evaluations, this normalized AUC equals the mean test

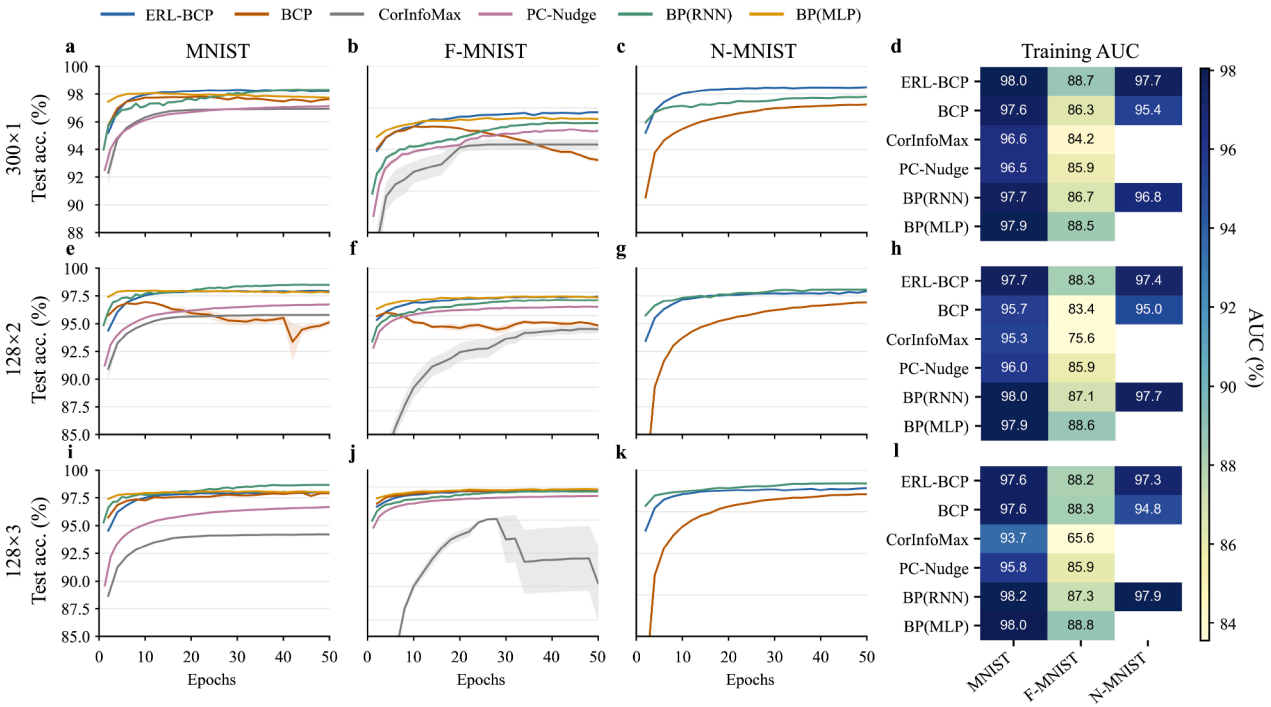


Fig. 2. Learning curves and normalized AUC values across datasets and network configurations. Rows correspond to the 300×1, 128×2, and 128×3 configurations, and columns correspond to MNIST, F-MNIST, and N-MNIST. Solid lines show the mean over independent random seeds, and shaded regions summarize seed variability. Heatmaps report the normalized area under the test-accuracy curve. Blank cells denote unavailable evaluations.

accuracy across the recorded training trajectory. Best accuracy summarizes peak performance, while AUC captures both early learning and sustained accuracy. Fig. 2 presents the learning curves and AUC heatmaps, and Tables 1 and 2 report mean best accuracy with 95% confidence intervals.

All task performance experiments use independent random initializations across runs. Uncertainty is reported as the 95% confidence interval of the mean (CI), calculated as $\mu \pm Z(\frac{\sigma}{\sqrt{n}})$, where μ and σ denote the sample mean and sample standard deviation across n independent runs, respectively. Z is the critical value corresponding to the 95% confidence level.

On MNIST, ERL-BCP attains the highest $\mathrm{Acc}_{\mathrm{best}}$ among the evaluated local methods in all three configurations. Its accuracies of 98.36%, 98.10%, and 98.14% exceed the strongest local baseline by 0.44, 1.08, and 0.06 percentage points, respectively. The ERL-BCP means span only 0.26 percentage points across configurations, compared with 1.06 for BCP, 0.48 for PC-Nudge, and 2.73 for CorInfoMax. Relative to the nonlocal references, ERL-BCP matches BP(RNN) under 300×1 and trails it by 0.44 and 0.59 percentage points under the two deeper configurations. Its differences from BP(MLP) are +0.25, +0.02, and −0.09 percentage points. ERL-BCP therefore provides the strongest local performance with the smallest variation across the tested MNIST configurations, while BP(RNN) remains higher in the two deeper settings.

ERL-BCP likewise achieves the highest $\mathrm{Acc}_{\mathrm{best}}$ among the local methods on F-MNIST. Its accuracies of 89.70%, 89.36%, and 89.38% exceed the strongest local baseline by 1.62, 2.39, and 0.20 percentage points. The three ERL-BCP means span 0.34 percentage points, whereas BCP decreases from 88.08% under 300×1 to 85.84% under 128×2 before recovering to 89.18% under 128×3. PC-Nudge and CorInfoMax remain lower in every configuration. ERL-BCP also exceeds BP(RNN) by 1.34, 0.89, and 0.65 percentage points, while its differences from BP(MLP) are +0.69, +0.05, and −0.18 percentage points. These results establish ERL-BCP as the strongest local method on F-MNIST with limited variation across the tested configurations.

Table 1
Best test accuracy on the static image benchmarks across network configurations. Results are reported as mean ± 95% confidence interval over 30 random seeds. Bold values indicate the highest mean accuracy for each dataset and configuration.

Method	Structure	MNIST ↑	Fashion -MNIST ↑
BP(MLP)	300×1	98.11 ± 0.037	89.01 ± 0.084
	128×2	98.08 ± 0.031	89.31 ± 0.098
	128×3	98.23 ± 0.045	$\mathbf{89.56 \pm 0.097}$
BP(RNN)	300×1	98.36 ± 0.050	88.36 ± 0.229
	128×2	$\mathbf{98.54 \pm 0.064}$	88.47 ± 0.115
	128×3	$\mathbf{98.73 \pm 0.057}$	88.73 ± 0.092
PC-Nudge	300×1	97.15 ± 0.072	87.55 ± 0.188
	128×2	96.74 ± 0.140	86.97 ± 0.254
	128×3	96.67 ± 0.132	87.24 ± 0.201
CorInfoMax	300×1	96.95 ± 0.063	85.64 ± 1.178
	128×2	95.80 ± 0.073	82.44 ± 1.783
	128×3	94.22 ± 0.061	82.80 ± 0.613
BCP	300×1	97.92 ± 0.054	88.08 ± 0.114
	128×2	97.02 ± 0.126	85.84 ± 0.217
	128×3	98.08 ± 0.043	89.18 ± 0.102
ERL-BCP	300×1	$\mathbf{98.36 \pm 0.023}$	$\mathbf{89.70 \pm 0.067}$
	128×2	98.10 ± 0.032	$\mathbf{89.36 \pm 0.058}$
	128×3	98.14 ± 0.048	89.38 ± 0.134

On N-MNIST, ERL-BCP reaches 98.16%, 97.97%, and 97.92% across the three configurations. Under 300×1, it achieves the highest overall accuracy, exceeding BP(RNN) by 0.88 percentage points and BCP by 1.58 percentage points. Under the 128×2 and 128×3 configurations, BP(RNN) outperforms ERL-BCP by 0.14 and 0.51 percentage points, respectively, whereas ERL-BCP maintains an advantage over BCP by 1.03 and 0.77 percentage points. Across the tested configurations, BP(RNN) and BCP increase by 1.15 and 0.57 percentage points from 300×1 to

Table 2
Best test accuracy across network configurations and benchmark tasks. Results are reported as mean ± 95% confidence interval over 30 random seeds. Bold values indicate the highest mean accuracy for each configuration.

Method	Structure	N-MNIST ↑	P-MNIST ↑	S-MNIST ↑
BP(RNN)	300×1	97.28±0.062	**97.51±0.053**	98.63±0.052
	128×2	**98.11±0.184**	97.10±0.085	98.77±0.039
	128×3	**98.43±0.039**	97.17±0.063	**98.87±0.033**
BCP	300×1	96.58±0.076	96.33±0.036	98.76±0.035
	128×2	96.94±0.066	97.33±0.050	98.71±0.072
	128×3	97.15±0.078	97.28±0.046	98.64±0.023
ERL-BCP	300×1	**98.16±0.031**	97.41±0.074	**98.85±0.050**
	128×2	97.97±0.043	**97.44±0.085**	**98.87±0.041**
	128×3	97.92±0.060	**97.34±0.092**	98.67±0.032

128×3, whereas ERL-BCP changes by -0.24 percentage points. ERL-BCP therefore provides the strongest result under 300×1 and consistently outperforms BCP, while BP(RNN) leads in the two deeper configurations.

On the two sequential benchmarks, ERL-BCP maintains competitive performance across all evaluated configurations. On P-MNIST, it achieves mean best accuracies of 97.41%, 97.44%, and 97.34% under the 300×1, 128×2, and 128×3 configurations, respectively, outperforming BCP in all three cases and remaining closely competitive with BP(RNN). On S-MNIST, ERL-BCP reaches 98.85%, 98.87%, and 98.67%, again exceeding BCP across all configurations. It also exceeds BP(RNN) under 300×1, matches it under 128×2, and remains within 0.20 percentage points under 128×3. Across configurations, the ERL-BCP mean varies by only 0.10 percentage points on P-MNIST and 0.20 percentage points on S-MNIST. These results extend the evaluation beyond static and event-based inputs and show that ERL-BCP retains stable predictive performance when the input is presented sequentially over time.

Across the five benchmarks, ERL-BCP maintains consistently high accuracy across the three evaluated network configurations, with only modest variations among different width-depth settings. The 300×1 configuration gives the highest ERL-BCP mean accuracy on MNIST, F-MNIST, and N-MNIST, whereas the 128×2 configuration gives the highest mean accuracy on P-MNIST and S-MNIST. Across the three configurations, the maximum variation in mean accuracy is only 0.26, 0.34, 0.24, 0.10, and 0.20 percentage points on MNIST, F-MNIST, N-MNIST, P-MNIST, and S-MNIST, respectively. These results show that the predictive performance of ERL-BCP is only weakly affected by the tested changes in network width and depth, while the optimal configuration remains task dependent.

AUC distinguishes training trajectories that peak accuracy alone does not capture. On MNIST, ERL-BCP has the highest local-method AUC under 300×1 and 128×2 and matches BCP under 128×3. On N-MNIST, its AUC values of 97.7%, 97.4%, and 97.3% exceed BCP by 2.3 to 2.5 percentage points, even though the corresponding best-accuracy margin narrows from 1.58 to 0.77 percentage points. The curves show that ERL-BCP reaches high accuracy earlier and sustains it, while BCP closes part of the peak-accuracy gap later in training. On F-MNIST, ERL-BCP also exceeds BCP in AUC under 300×1 and 128×2, with 88.7% versus 86.3% and 88.3% versus 83.4%. Their AUC values are nearly equal under 128×3 at 88.2% and 88.3%, indicating configuration-dependent differences in the training trajectories. CorInfoMax is unstable under the 128×3 F-MNIST configuration, where its AUC falls to 65.6% and seed variability increases substantially.

4.2. Direct dynamical characterization of the energy-regulated liquid timescale

Task-level performance establishes the utility of the proposed dynamics for learning, but it does not directly characterize how the effective liquid timescale responds to the internal operating state or how the timescale determines membrane-state evolution. We therefore use controlled dynamical probes to examine the two causal links underlying the ERL dynamics: the mapping from current load to the effective timescale and the mapping from the effective timescale to the rate of state evolution.

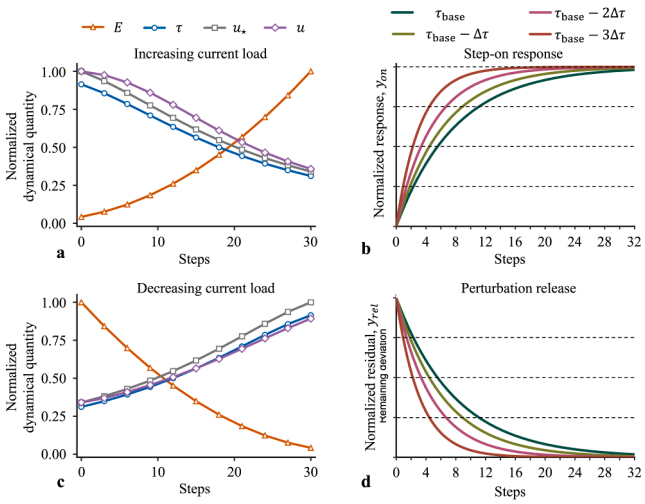


Fig. 3. Direct characterization of the energy-regulated liquid timescale. (a,c) Increasing- and decreasing-current-load probes under a fixed signed net drive, with q updated at Step $= 0, 3, \ldots, 30$. Changes in current load regulate E and τ, while $u_\star$ and u show the corresponding target and state responses. (b,d) Normalized step-on and perturbation-release responses under four frozen timescales. Shorter τ produces faster state establishment and perturbation recovery. Quantities in (a) and (c) are normalized by their respective reference values, and each response in (b) and (d) is normalized by its own amplitude.

To isolate the first relation from changes in signed membrane drive, the current-load probes introduce a common-mode current component q,

$$I_{\text{exc}} = q + D_0, \qquad I_{\text{inh1}} = q, \qquad I_{\text{inh2}} = 0, \tag{22}$$

so that

$$D = I_{\text{exc}} - I_{\text{inh1}} - I_{\text{inh2}} = D_0 \tag{23}$$

remains constant, whereas the quadratic current load varies as

$$E = (q + D_0)^2 + q^2. \tag{24}$$

This construction changes the absolute current load while preserving the signed net membrane drive. The condition $I_{\text{inh2}} = 0$ is used only for this controlled probe and does not represent the full ERL-BCP network. The corresponding regulator follows

$$\lambda = \frac{1}{\tau_{\text{base}}} + \mu E, \qquad \tau = \frac{1}{\lambda}, \qquad u_\star = D_0 \tau. \tag{25}$$

The complete probe parameters and piecewise-constant state solution are provided in Appendix D.

Fig. 3(a,c) shows the resulting current-load responses. Increasing the common-mode current component q raises E and progressively shortens τ, whereas reducing the current load allows τ to increase accordingly. Because $u_\star = D_0 \tau$, the moving equilibrium varies with the regulated timescale, and the membrane state tracks this equilibrium with a finite temporal lag. Importantly, D_0 remains unchanged throughout these probes, indicating that the modulation of τ is driven by changes in the absolute current load rather than by changes in the signed net drive.

The temporal consequence of the timescale is examined separately in Fig. 3(b,d) using four frozen normalized timescales, $\tau/\tau_{\text{base}} \in \{1.0, 0.8, 0.6, 0.4\}$. The step-on responses show that trajectories with

Table 3
Mechanism configurations of ERL-BCP, its four controlled variants, and the BCP reference method. Within the ERL dynamics, Dynamic Gain denotes the $\tau(t)$-dependent scaling of the instantaneous membrane target.

Method	Configuration	E/I Balance Signal	ERL Dynamics: ERL Timescale	ERL Dynamics: Dynamic Gain	Bounded Feedback
ERL-BCP	ERL-BCP	✓	✓	✓	✓
	Generic local error	×	✓	✓	✓
	Gain-decoupled	✓	✓	×	✓
	Fixed-timescale	✓	×	×	✓
	Unbounded feedback	✓	✓	✓	×
BCP	BCP	✓	×	×	×

Table 4
Comparison of ERL-BCP, its four controlled mechanism variants, and BCP. Results are reported as mean ± standard deviation over five seed-aligned runs. Balance residual is the mean absolute closed-loop E/I current mismatch, and current energy is the mean closed-loop assembly current energy.

Dataset	Method	Test Acc. (%) ↑	Test Loss ↓	Balance Residual $\mathcal{R}_{\text{bal}}$ ↓	Current Energy E_{cur} ↓
MNIST	ERL-BCP	**98.36 ± 0.02**	**0.065 ± 0.001**	**0.253 ± 0.009**	**3.24 ± 0.18**
	Generic local error	96.84 ± 0.32	0.204 ± 0.022	7.651 ± 0.572	$(9.10 \pm 0.77) \times 10^3$
	Gain-decoupled	98.22 ± 0.03	0.073 ± 0.002	0.262 ± 0.010	4.10 ± 0.35
	Fixed-timescale	98.05 ± 0.03	0.090 ± 0.002	0.313 ± 0.005	7.06 ± 0.07
	Unbounded feedback	98.27 ± 0.05	0.069 ± 0.001	0.447 ± 0.053	10.7 ± 1.07
	BCP	97.92 ± 0.05	0.095 ± 0.010	0.510 ± 0.072	14.9 ± 3.30
F-MNIST	ERL-BCP	**89.70 ± 0.07**	**0.336 ± 0.006**	**0.354 ± 0.031**	**6.71 ± 0.89**
	Generic local error	80.20 ± 1.57	2.508 ± 1.246	12.24 ± 6.765	$(2.82 \pm 2.01) \times 10^4$
	Gain-decoupled	89.18 ± 0.18	0.357 ± 0.011	0.392 ± 0.031	7.42 ± 1.01
	Fixed-timescale	88.54 ± 0.15	0.382 ± 0.012	0.430 ± 0.032	7.44 ± 1.07
	Unbounded feedback	89.35 ± 0.31	0.344 ± 0.057	0.949 ± 0.029	30.6 ± 0.97
	BCP	88.08 ± 0.11	0.392 ± 0.043	1.109 ± 0.104	46.7 ± 8.85

smaller τ reach the same normalized fraction of the target state earlier. The perturbation-release responses show the complementary effect: smaller τ contracts deviations from equilibrium more rapidly. Together, these probes directly characterize the dynamical chain implemented by the proposed regulator: increasing current load raises the decay rate and shortens the effective liquid timescale, while a shorter timescale accelerates membrane-state establishment and perturbation recovery.

4.3. ERL-BCP mechanisms support accurate and regulated local learning

We examine how the controlled components summarized in Table 3 contribute to predictive performance and to a balanced, low-current closed-loop operating state. ERL-BCP is the full model and combines the balance-derived signal, energy-regulated liquid timescale, and bounded inhibitory feedback. For controlled analysis, the ERL dynamics are further separated into the ERL Timescale and Dynamic Gain. Each controlled variant targets one designated component while leaving the remaining formulation unchanged.

The open-loop state is obtained with task feedback disabled, and the closed-loop state is obtained with task feedback enabled. The generic local error variant replaces the balance-derived signal with the difference between open-loop and closed-loop excitatory activity:

$$\delta_i^{(a)} = r_{\text{exc},i}^{\text{OL},(a)} - r_{\text{exc},i}^{\text{CL},(a)}. \tag{26}$$

It retains both the energy-regulated liquid timescales and bounded inhibitory feedback.

The gain-decoupled variant retains the ERL Timescale while removing Dynamic Gain:

$$\frac{du_i^{E,(a)}(t)}{dt} = \lambda^{(a)}(t)\left[-u_i^{E,(a)}(t) + \tau_{\text{base}} D_i^{(a)}(t)\right]. \tag{27}$$

Accordingly, $\tau^{(a)}(t) = 1/\lambda^{(a)}(t)$ remains adaptive, while $u_{i,\star}^{(a)}(t) = \tau_{\text{base}} D_i^{(a)}(t)$.

The fixed-timescale variant further removes adaptive temporal relaxation by replacing $\lambda^{(a)}(t)$ with the constant $\lambda_{\text{fixed}} = \frac{1}{\tau_{\text{fixed}}}$, $\tau_{\text{fixed}} = \bar{\tau}_{\text{ERL}} = 0.0271$, where $\bar{\tau}_{\text{ERL}}$ is the measured mean effective timescale over the adaptive ERL trajectories. Its dynamics are therefore

$$\frac{du_i^{E,(a)}(t)}{dt} = \lambda_{\text{fixed}}\left[-u_i^{E,(a)}(t) + \tau_{\text{base}} D_i^{(a)}(t)\right]. \tag{28}$$

This matched fixed-timescale setting removes both the ERL Timescale and Dynamic Gain while preserving the remaining formulation. A broader sensitivity analysis over five fixed-timescales, including τ_{base}, is provided in Appendix C.

The unbounded feedback variant retains the balance-derived signal and energy-regulated liquid timescales while removing the bound on inhibitory feedback. BCP retains the balance-derived teaching signal but includes neither of the two proposed regulatory mechanisms. It is therefore treated as an independent method-level reference rather than as a single-mechanism ablation.

All methods are evaluated on MNIST and F-MNIST with the same five random seeds, enabling seed-aligned comparisons. Test accuracy and loss quantify predictive performance, while balance residual and current energy characterize the closed-loop hidden-layer operating state. The two groups of metrics are analyzed separately because a regulatory mechanism can substantially constrain internal currents while producing only a small change in classification accuracy.

Let $\Omega_\Delta^{\text{CL}}$ denote the set of all recorded assembly-level balance deviations during closed-loop evaluation. The mean absolute balance residual is defined as

$$\mathcal{R}_{\text{bal}} = \frac{1}{\left|\Omega_\Delta^{\text{CL}}\right|} \sum_{\Delta \in \Omega_\Delta^{\text{CL}}} |\Delta|. \tag{29}$$

Likewise, let Ω_E^{CL} denote the corresponding set of recorded assembly-level current energies. The mean current energy is defined as

$$E_{\text{cur}} = \frac{1}{\left|\Omega_E^{\text{CL}}\right|} \sum_{E \in \Omega_E^{\text{CL}}} E. \tag{30}$$

Lower values of $\mathcal{R}_{\text{bal}}$ and E_{cur} indicate a more closely balanced state and a smaller overall current magnitude, respectively.

Table 4 shows that the complete ERL-BCP configuration provides the strongest overall combination of predictive performance and closed-loop dynamical regulation. ERL-BCP achieves the highest mean test accuracy on both datasets, with $98.36 \pm 0.02\%$ on MNIST and $89.70 \pm 0.07\%$ on F-MNIST. It also obtains the lowest mean test loss, balance residual, and current energy on both datasets. These results indicate that the full configuration provides the most favorable operating point among the evaluated methods, combining accurate classification with the closest closed-loop E/I balance and the lowest current energy.

Replacing the balance-derived signal with the generic local error causes the largest degradation among the controlled variants. Relative to ERL-BCP, mean test accuracy decreases by 1.52 percentage points on MNIST and 9.50 percentage points on F-MNIST, with markedly greater seed variability on F-MNIST. The deterioration in the dynamical metrics is substantially larger. The balance residual increases by factors of 30.2 and 34.6, while the current energy increases by factors of 2809 and 4203 on MNIST and F-MNIST, respectively. Because this controlled variant retains the energy-regulated liquid timescales and bounded inhibitory feedback, the observed differences can be attributed to replacing the balance-derived signal within the evaluated configuration. The open-loop and closed-loop activity contrast still supports nontrivial classification performance, although it provides a substantially less effective local learning signal and fails to preserve a regulated closed-loop operating state. The balance-derived signal therefore plays the central role in coupling local synaptic updates to both task performance and the internal E/I operating condition.

The gain-decoupled variant achieves higher mean accuracy than the fixed-timescale variant on both datasets. On MNIST, mean accuracy increases from 98.05% to 98.22%, while test loss decreases from 0.090 to 0.073 and the balance residual decreases from 0.313 to 0.262. On F-MNIST, mean accuracy increases from 88.54% to 89.18%, accompanied by reductions in test loss from 0.382 to 0.357 and in balance residual from 0.430 to 0.392. Thus, the ERL Timescale retains an advantage in predictive performance and balance regulation after Dynamic Gain is removed, with a larger accuracy difference on F-MNIST.

ERL-BCP further improves upon the gain-decoupled variant, increasing mean accuracy by 0.14 percentage points on MNIST and 0.52 percentage points on F-MNIST, together with further reductions in test loss and balance residual. The current-energy results distinguish the two effects more clearly. On MNIST, E_{cur} decreases from 7.06 for the fixed-timescale variant to 4.10 for the gain-decoupled variant and to 3.24 for ERL-BCP. On F-MNIST, the fixed-timescale and gain-decoupled variants remain nearly identical at 7.44 and 7.42, whereas ERL-BCP reduces E_{cur} to 6.71. These results indicate that the ERL Timescale contributes independently to predictive performance and balance regulation, while Dynamic Gain provides an additional, task-dependent contribution to the full ERL dynamics.

We further examine whether this comparison depends on the selected constant timescale. Five fixed values spanning the admissible range are evaluated in Appendix C, including $\tau_{\mathrm{fixed}} = 0.0271$, which matches the measured mean effective timescale of the adaptive ERL trajectories. This setting also provides the strongest fixed-timescale result among the evaluated constants, reaching 98.05% on MNIST and 88.54% on F-MNIST. ERL-BCP nevertheless reaches 98.36% and 89.70%, corresponding to improvements of 0.31 and 1.16 percentage points, respectively. Thus, the advantage of adaptive temporal regulation persists both when the fixed reference is matched to the empirical mean operating timescale and across the broader range of evaluated fixed-timescales.

The unbounded feedback variant exhibits a distinct separation between test performance and dynamical regulation. Removing the feedback bound changes mean accuracy by only 0.09 percentage points on MNIST and 0.35 percentage points on F-MNIST. Its test loss remains close to that of ERL-BCP on both datasets, although the F-MNIST result shows substantially greater variability. These small result differences contrast with substantial changes in the internal operating state. The balance residual increases by factors of 1.8 and 2.7, and the current energy increases by factors of 3.3 and 4.6 on MNIST and F-MNIST, respectively. This dissociation supports the intended role of bounded inhibitory feedback as a dynamical safeguard: it constrains feedback-driven current amplification and reduces closed-loop E/I mismatch while largely preserving task performance.

Table 5
Paired update-alignment statistics. Cosine alignments and paired advantages are reported as mean ± standard deviation over 155 matched checkpoint and probe-batch evaluations from one run per dataset. Positive pairs report the number and proportion of evaluations satisfying $A_{\mathrm{ERL}} > A_{\mathrm{GEN}}$.

Metric	MNIST	F-MNIST
ERL-BCP alignment A_{ERL}	0.7043 ± 0.0976	0.6940 ± 0.1104
Generic local alignment A_{GEN}	0.5889 ± 0.1026	0.5859 ± 0.1250
Paired advantage ΔA	0.1155 ± 0.0303	0.1081 ± 0.0267
Positive pairs, $\Delta A > 0$	155/155 (100%)	155/155 (100%)

BCP provides an independent balance-based reference for assessing the combined contribution of the two proposed regulatory mechanisms. Its test loss is higher on both datasets. The differences are more pronounced in the dynamical metrics: compared with ERL-BCP, BCP shows approximately 2.0 and 3.1 times higher balance residuals, together with approximately 4.6 and 7.0 times higher current energies on MNIST and F-MNIST, respectively. Because BCP differs from ERL-BCP in both regulatory mechanisms, this comparison provides method-level evidence.

Its results nevertheless agree with the controlled variants and support the complementary roles of the proposed mechanisms. The gain-decoupled comparison further shows that the ERL Timescale retains advantages in predictive performance and balance regulation after Dynamic Gain is removed, while Dynamic Gain provides additional regulation within the full ERL dynamics. Bounded inhibitory feedback provides complementary protection against excessive current growth. Combined with the balance-derived signal, these mechanisms enable ERL-BCP to achieve the most favorable overall operating point, characterized by high and consistent test performance, low E/I mismatch, and low internal current energy.

4.4. ERL-BCP provides consistently task-aligned local updates

The preceding experiments establish the predictive performance, dynamical timescale behavior, and mechanism-level contributions of ERL-BCP. However, these results do not determine whether an individual local parameter update follows a task-relevant direction at a particular point during training. We therefore use the supervised backpropagation (BP) descent direction as a reference and compare it with two local update directions under matched conditions (Fig. 4(a)). The ERL-BCP direction is generated from the balance-derived signal used by the proposed framework. The generic local error variant retains the same parameter state and probe batch while replacing balance-derived signal with the difference between open-loop and closed-loop excitatory activity. This controlled replacement tests whether the E/I balance-derived signal contributes task-aligned directional information beyond that contained in a generic local activity contrast.

For each dataset, one training trajectory of 30 epochs was sampled at initialization and after every epoch, producing 31 parameter checkpoints. At each checkpoint, the ERL-BCP, generic local error, and BP directions were computed from the same fixed parameter state on the same five probe batches. This procedure yields 155 pairedf evaluations per dataset and controls for parameter location, sample composition, and training stage within every comparison. Because the analysis follows one run per dataset, the reported standard deviations describe variation across checkpoints and probe batches within the sampled trajectory.

Let $\mathbf{d}_s^{k,b}$ denote the parameter-update direction produced by rule s at checkpoint k on probe batch b, using a common sign convention in

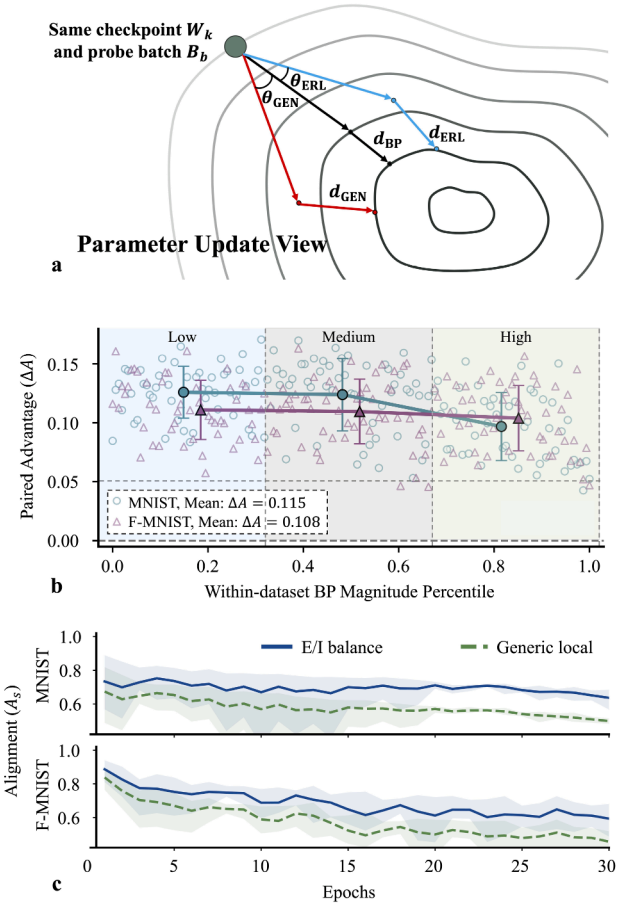


Fig. 4. Directional alignment of local update directions with supervised descent. (a) Conceptual geometry of the comparison. The BP descent direction provides the supervised reference, and each local update is evaluated according to its angular alignment with this reference. The illustrated paths are schematic. (b) Paired alignment advantage $\Delta A = A_{\mathrm{ERL}} - A_{\mathrm{GEN}}$ plotted against the within-dataset BP magnitude percentile for all 155 matched checkpoint and probe-batch evaluations on each dataset. Each small marker represents one paired evaluation. The horizontal axis ranks the BP reference-direction norm within each dataset and maps it to the interval $[0, 1]$. The shaded regions denote low, medium, and high BP magnitude percentile intervals. Enlarged markers and error bars show the mean ± standard deviation of ΔA within each interval, and connecting lines summarize the binned trend. Points above $\Delta A = 0$ favor ERL-BCP. (c) Epoch-wise BP alignment of ERL-BCP and generic local update directions. Curves show the mean over five fixed probe batches at each checkpoint, and shaded bands indicate the corresponding standard deviation across probe batches.

which $\mathbf{d}_{\mathrm{BP}}^{k,b}$ denotes the supervised descent direction. The alignment with BP is defined as

$$A_s^{k,b} = \frac{\left\langle \mathbf{d}_s^{k,b}, \mathbf{d}_{\mathrm{BP}}^{k,b} \right\rangle}{\left\| \mathbf{d}_s^{k,b} \right\|_2 \left\| \mathbf{d}_{\mathrm{BP}}^{k,b} \right\|_2}, \qquad s \in \{\mathrm{ERL}, \mathrm{GEN}\}, \tag{31}$$

where ERL denotes the balance-derived local rule used by ERL-BCP and GEN denotes the matched generic local error rule. Their paired alignment difference is

$$\Delta A^{k,b} = A_{\mathrm{ERL}}^{k,b} - A_{\mathrm{GEN}}^{k,b}. \tag{32}$$

Cosine normalization isolates directional agreement from differences in update magnitude. A positive $\Delta A^{k,b}$ indicates that the ERL-BCP update has a larger normalized component along supervised descent direction than the matched generic local update at the same checkpoint and on the same probe batch.

To analyze whether this advantage depends on the strength of the supervised reference update, Fig. 4(b) ranks the BP reference-direction norm within each dataset. For the i-th paired evaluation in a dataset with N evaluations, the percentile coordinate is

$$p_i = \frac{r_i}{N-1}, \tag{33}$$

where r_i is the within-dataset rank of $\left\| \mathbf{d}_{\mathrm{BP}}^{(i)} \right\|_2$. Thus, $p_i = 0$ corresponds to the smallest BP-direction norm in that dataset, $p_i = 1$ corresponds to the largest, and intermediate values indicate relative BP-update strength within the same dataset. This within-dataset normalization prevents differences in the absolute scale of BP-direction norms between MNIST and F-MNIST from dominating the comparison.

Both local rules exhibit positive mean cosine similarity with BP, indicating that the generic local activity contrast also contains task-relevant directional information. ERL-BCP nevertheless shows consistently stronger alignment. Its mean alignment reaches 0.7043 on MNIST and 0.6940 on F-MNIST, compared with 0.5889 and 0.5859 for the generic local error variant, respectively (Table 5). The corresponding paired advantages are 0.1155 ± 0.0303 and 0.1081 ± 0.0267, and every paired evaluation satisfies $\Delta A > 0$ on both datasets.

Fig. 4(b) further examines whether the paired alignment advantage changes with the relative strength of the supervised reference direction. Across MNIST and the more challenging F-MNIST task, the mean ΔA remains clearly above zero in the low, medium, and high BP magnitude percentile regions, showing that ERL-BCP consistently provides a more BP-aligned local direction than the generic local error rule across both datasets and across different BP-magnitude regimes. The binned means change little from the low to medium regions and decrease only mildly in the high-percentile region, while remaining positive. This pattern indicates that the advantage of the balance-derived ERL-BCP teaching signal is preserved across weak, moderate, and strong supervised reference directions. Thus, the directional benefit of ERL-BCP is broadly distributed across the sampled parameter states and probe batches, rather than being confined to a narrow BP magnitude regime.

Fig. 4(c) further evaluates whether the directional advantage of ERL-BCP is preserved across training stages. On MNIST, the mean ΔA is 0.0747 for checkpoints 0–5 and increases to 0.1388 for checkpoints 16–30. On F-MNIST, it similarly increases from 0.0702 to 0.1261. Although the absolute BP alignment of both local directions generally decreases as training proceeds, the ERL-BCP curve remains above the generic local error curve at every checkpoint. This separation becomes more pronounced in the later stages, especially on F-MNIST, with a larger decline observed for the generic local error direction, particularly on F-MNIST. Although the standard-deviation bands overlap at some checkpoints, the paired comparison remains consistent because every checkpoint and probe-batch evaluation yields $\Delta A > 0$. These results show that the balance-derived ERL-BCP signal provides a stable directional advantage from early to late training, supporting its role as a task-aligned local teaching signal under changing optimization conditions.

Together, the aggregate statistics, BP magnitude percentile analysis, and temporal profiles support the role of the balance-derived signal in task-aligned local learning. At identical parameter states and on identical probe batches, this signal consistently reorients the ERL-BCP update toward supervised descent relative to the matched generic local activity contrast. This result directly supports the design objective of ERL-BCP: to convert local E/I balance regulation into a more task-aligned teaching signal while preserving the locality of the update.

5. Discussion and conclusion

ERL-BCP couples local E/I balance-controlled plasticity with state-dependent continuous-time regulation. Its central dynamical feature is that the operating condition of each assembly modulates the decay rate of the excitatory membrane state. Increasing the current-energy

measure raises $\lambda(t)$ and shortens the effective timescale, thereby accelerating relaxation under high-current conditions. The controlled current-load probes further show that this regulation is effective: increasing absolute current load shortens τ, whereas decreasing the load restores a longer timescale while the signed net membrane drive remains fixed. Complementary timescale-response probes show that shorter timescales accelerate both state establishment and perturbation recovery. This monotonic mapping is adopted as a computational regulatory principle motivated by conductance-dependent neuronal dynamics, in which increased synaptic conductance is associated with a shorter effective membrane time constant.

The ERL dynamics inherently couple temporal adaptation with the instantaneous drive-to-state gain, because changes in the effective timescale influence both how rapidly the membrane state evolves and the state toward which it relaxes. The gain-decoupled analysis helps distinguish these two effects by preserving adaptive temporal regulation while removing the timescale-dependent modulation of the instantaneous target. The resulting comparison indicates that adaptive temporal regulation has a functional role beyond simple rescaling of the membrane target. Dynamic Gain then provides an additional mechanism through which the current operating condition can reshape the state response of the assembly. From this perspective, the two effects are complementary: adaptive temporal relaxation governs how quickly the local circuit responds to changing conditions, whereas Dynamic Gain influences the amplitude of the corresponding state adjustment. Their coupling therefore allows ERL-BCP to regulate both the temporal and state-response characteristics of the local E/I dynamics.

Inhibition plays two complementary roles in ERL-BCP. Local E/I deviations provide the balance-derived signal for synaptic plasticity, while the balance-sensitive inhibitory pathway conveys task-dependent perturbations to the local circuit. Bounding this feedback produces relatively small changes in classification accuracy but substantially reduces closed-loop current energy and balance residual, indicating that its primary role lies in regulating the internal operating regime. Directional-alignment analysis provides complementary evidence at the update level: within the sampled training trajectories, the balance-derived update remains more closely aligned with supervised descent than the matched generic local error under all paired evaluations.

Across static, event-based, and sequential benchmarks, ERL-BCP maintains strong local-learning performance while regulating the continuous-time dynamics from which its local plasticity signal is generated. The present study is limited to rate-based networks, moderate-scale classification tasks, and one sampled alignment trajectory per dataset. Further evaluation on larger temporal problems, multiple independent alignment trajectories, and event-driven spiking implementations is required to establish the generality of these mechanisms.

CRediT authorship contribution statement

Yichuan Jiang: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization; **Qian Zhao:** Validation, Investigation, Data curation; **Yusheng Yang:** Writing – review & editing, Supervision, Funding acquisition; **Yangmin Xie:** Writing – review & editing, Validation, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

Appendix A. Uniform boundedness of the energy-regulated membrane dynamics

In this appendix, we establish uniform boundedness of the excitatory membrane dynamics under the energy-regulated liquid timescales introduced in the main text. The analysis is carried out for the Softplus and then extended to the Tanh case as a robustness result.

We denote by a the index of a local assembly and by i the index of an excitatory neuron within that assembly. External input variables are denoted by $x_j(t)$, latent membrane potentials by $u_i^{(a)}(t)$, and firing rates by $r_i^{(a)}(t)$. The three current components $I_i^{\mathrm{exc},(a)}(t)$, $I_i^{\mathrm{inh1},(a)}(t)$, and $I_i^{\mathrm{inh2},(a)}(t)$, together with the net driving current $D_i^{(a)}(t)$, the assembly-level energy $E^{(a)}(t)$, the decay rate $\lambda^{(a)}(t)$, and the instantaneous target $u_{i,\star}^{(a)}(t)$, are defined in the main text.

Throughout the analysis, we assume that the external input is bounded and piecewise continuous, that all weight matrices and feedback mappings are uniformly bounded, and that the induced vector field is locally Lipschitz with respect to the state variables. These assumptions ensure existence and uniqueness of solutions on finite time intervals.

Consistent with all experiments reported in this study, the boundedness analysis below assumes $R_\nu = 1$. Under this setting, the assembly-level current-energy measure in Eq. (9) reduces to the unweighted sum of squared pathway currents. For the proof, it is convenient to introduce the neuron-wise local current-energy contribution

$$E_{i,\mathrm{loc}}^{(a)}(t) = \left(I_i^{\mathrm{exc},(a)}(t)\right)^2 + \left(I_i^{\mathrm{inh1},(a)}(t)\right)^2 + \left(I_i^{\mathrm{inh2},(a)}(t)\right)^2 \tag{A.1}$$

By construction, this local contribution satisfies $E_{i,\mathrm{loc}}^{(a)}(t) \le E^{(a)}(t)$.

Our goal is to show that the energy-regulated liquid timescales induces an absolute bound on the instantaneous target $u_{i,\star}^{(a)}(t)$, which in turn yields a forward-invariant interval for the membrane potential trajectory $u_i^{(a)}(t)$.

A.1. Softplus activation

Under the Softplus activation adopted in the main model, firing rates are nonnegative. Consequently, the three current components preserve the intended excitatory-inhibitory sign structure, and the assembly-level energy remains nonnegative for all t. It follows immediately that the decay rate is strictly positive, since $\lambda^{(a)}(t) = 1/\tau_{\mathrm{base}} + \mu E^{(a)}(t) \ge 1/\tau_{\mathrm{base}} > 0$.

A.1.1. Absolute bound on the instantaneous target

For every excitatory neuron i in every assembly a, the instantaneous target state satisfies

$$\left|u_{i,\star}^{(a)}(t)\right| \le M_{\mathrm{abs}}, \qquad M_{\mathrm{abs}} = \sqrt{\frac{3\tau_{\mathrm{base}}}{4\mu}}, \qquad \forall t \ge 0. \tag{A.2}$$

Proof. Using the Cauchy–Schwarz inequality, we obtain

$$\begin{aligned}\left(D_i^{(a)}(t)\right)^2 &= \left(I_i^{\mathrm{exc},(a)}(t) - I_i^{\mathrm{inh1},(a)}(t) - I_i^{\mathrm{inh2},(a)}(t)\right)^2 \\ &\le \left(1^2 + (-1)^2 + (-1)^2\right)\Big[\left(I_i^{\mathrm{exc},(a)}(t)\right)^2 \\ &\quad + \left(I_i^{\mathrm{inh1},(a)}(t)\right)^2 + \left(I_i^{\mathrm{inh2},(a)}(t)\right)^2\Big] \\ &= 3E_{i,\mathrm{loc}}^{(a)}(t) \le 3E^{(a)}(t)\end{aligned} \tag{A.3}$$

Substituting this estimate into the expression for $u_{i,\star}^{(a)}(t)$ yields

$$\left|u_{i,\star}^{(a)}(t)\right| \le \frac{\left|D_i^{(a)}(t)\right|}{\frac{1}{\tau_{\mathrm{base}}} + \frac{\mu}{3}\left(D_i^{(a)}(t)\right)^2}$$

Applying the arithmetic–geometric mean inequality $a+b \geq 2\sqrt{ab}$ with $a = 1/\tau_{\text{base}}, b = \mu\big(D_i^{(a)}(t)\big)^2/3$, we obtain

$$\left|u_{i,\star}^{(a)}(t)\right| \leq \frac{\left|D_i^{(a)}(t)\right|}{2\sqrt{\frac{1}{\tau_{\text{base}}}\cdot\frac{\mu}{3}\big(D_i^{(a)}(t)\big)^2}} = \sqrt{\frac{3\tau_{\text{base}}}{4\mu}}$$

This proves (A.2). The bound in (A.2) is a global worst-case bound. Its purpose is to establish absolute boundedness of the state trajectory under the energy-dependent dynamics. It is not intended to characterize the typical operating range of membrane potentials during ordinary training or inference.

A.1.2. Uniform boundedness of the membrane potential

Define $U_i^{(a)} := \max\left\{\left|u_i^{(a)}(0)\right|, M_{\text{abs}}\right\}$. Then the membrane potential satisfies

$$\left|u_i^{(a)}(t)\right| \leq U_i^{(a)}, \qquad \forall t \geq 0. \tag{A.4}$$

Proof. We show that the interval $\left[-U_i^{(a)}, U_i^{(a)}\right]$ is forward invariant.

Suppose first that $u_i^{(a)}(t) = U_i^{(a)}$. Since $\left|u_{i,\star}^{(a)}(t)\right| \leq M_{\text{abs}} \leq U_i^{(a)}$, it follows that $u_{i,\star}^{(a)}(t) \leq U_i^{(a)}$. Writing the dynamics in centered form,

$$\frac{du_i^{(a)}(t)}{dt} = -\lambda^{(a)}(t)\big(u_i^{(a)}(t) - u_{i,\star}^{(a)}(t)\big)$$

we obtain

$$\frac{du_i^{(a)}(t)}{dt} = -\lambda^{(a)}(t)\big(U_i^{(a)} - u_{i,\star}^{(a)}(t)\big) \leq 0$$

because $\lambda^{(a)}(t) > 0$. Hence, the vector field points inward at the upper boundary.

Similarly, if $u_i^{(a)}(t) = -U_i^{(a)}$, then $u_{i,\star}^{(a)}(t) \geq -U_i^{(a)}$, and therefore

$$\frac{du_i^{(a)}(t)}{dt} = -\lambda^{(a)}(t)\big(-U_i^{(a)} - u_{i,\star}^{(a)}(t)\big) \geq 0$$

Thus, the vector field also points inward at the lower boundary.

Since the trajectory cannot cross either boundary outward, the interval $\left[-U_i^{(a)}, U_i^{(a)}\right]$ is forward invariant. This proves (A.4).

Taken together, the preceding two propositions show that under Softplus activation, the assembly-level energy-regulated liquid timescale induces an absolute bound on the instantaneous target and thereby induces uniform boundedness under the stated assumptions of the membrane trajectory.

A.2. Extension to Tanh activation

The same boundedness mechanism extends to the Tanh activation, which shows that the energy-regulated mechanism is not tied to a specific nonnegative activation function.

Under Tanh activation, the current components may take negative values. Nevertheless, the assembly-level energy remains nonnegative by construction, and hence the decay rate is still strictly positive: $\lambda^{(a)}(t) = 1/\tau_{\text{base}} + \mu E^{(a)}(t) \geq 1/\tau_{\text{base}} > 0$. Moreover, the estimate $\big(D_i^{(a)}(t)\big)^2 \leq 3E_{i,\text{loc}}^{(a)}(t) \leq 3E^{(a)}(t)$ remains valid regardless of the signs of the current components. Therefore, the same absolute bound follows:

$$\left|u_{i,\star}^{(a)}(t)\right| \leq \sqrt{\frac{3\tau_{\text{base}}}{4\mu}}, \qquad \forall t \geq 0. \tag{A.5}$$

Applying the same forward-invariance argument yields

$$\left|u_i^{(a)}(t)\right| \leq \max\left\{\left|u_i^{(a)}(0)\right|, \sqrt{\frac{3\tau_{\text{base}}}{4\mu}}\right\}, \qquad \forall t \geq 0. \tag{A.6}$$

This extension shows that the boundedness result is structurally robust under both Softplus and Tanh activations. The two cases differ, however, in interpretation. Under Softplus activation, the nonnegativity of current components is naturally aligned with the sign-constrained excitatory-inhibitory decomposition adopted in the main model. Under Tanh activation, boundedness is preserved mathematically, but negative-valued channel outputs weaken the biological interpretability of fixed excitatory and inhibitory roles. The Tanh case should therefore be viewed as a generalization result demonstrating the broader applicability of the quadratic energy mechanism.

A.3. Boundedness of the effective time constant

Finally, the effective time constant defined in the main text can be written as

$$\tau^{(a)}(t) = \frac{\tau_{\text{base}}}{1 + \mu\tau_{\text{base}}E^{(a)}(t)} \tag{A.7}$$

Since $E^{(a)}(t) \geq 0$, it follows directly that

$$0 < \tau^{(a)}(t) \leq \tau_{\text{base}} \tag{A.8}$$

Thus, the effective time constant remains strictly positive and uniformly upper bounded throughout the evolution. Higher assembly-level current energy shortens the effective timescale and accelerates relaxation toward the instantaneous target, while preserving the derived bound on the excitatory membrane dynamics.

Appendix B. Variable conventions and hyperparameter settings

Table B.6
Hyperparameter settings used in the ERL-BCP experiments.

Category	Parameter	Value
ERL	τ_{base}	0.03
ERL	μ	0.6
ERL	$R_{\text{exc}}, R_{\text{inh1}}, R_{\text{inh2}}$	1
Balance	α	0.5
Controller	K_p	0.05
Controller	K_i	0.1
Controller	L_c	5
Optimization	Optimizer	Adam
Optimization	Learning Rate	5×10^{-4}
Training	Batch Size	64
Training	Epochs	50
Solver	Solver Method	Euler
Solver	Δt	0.01
Solver	T	1.0

For completeness and reproducibility, this section summarizes the variable conventions and implementation settings used in the ERL-BCP formulation and experiments. All state variables, rate outputs, current-like quantities, balance variables, temporal variables, decay rates, and the assembly-level current-energy measure are represented as dimensionless computational quantities.

Table B.6 reports the principal hyperparameters used for the network dynamics, E/I balance mechanism, bounded inhibitory feedback, optimization procedure, and numerical integration. Unless otherwise specified, these values are kept fixed across the reported experiments. The continuous-time dynamics are integrated using a fixed-step forward Euler method.

Appendix C. Sensitivity to the fixed membrane timescale

To examine whether the performance of the fixed-timescale variant depends on the selected constant value, we further evaluate five fixed excitatory membrane timescales on MNIST and F-MNIST. For the adaptive ERL dynamics, the effective timescale satisfies

$$0 < \tau^{(a)}(t) \leq \tau_{\text{base}}, \tag{C.1}$$

Table C.7
Classification accuracy under different fixed excitatory membrane timescales and adaptive ERL-BCP. Results are reported as mean ± standard deviation over five seed-aligned runs.

Timescale Setting	MNIST (%)	F-MNIST (%)
$\tau_{\text{fixed}} = 0.0075$	97.20 ± 0.13	84.80 ± 0.58
$\tau_{\text{fixed}} = 0.0150$	97.68 ± 0.08	86.90 ± 0.34
$\tau_{\text{fixed}} = 0.0225$	97.96 ± 0.05	88.10 ± 0.22
$\tau_{\text{fixed}} = 0.0271$	98.05 ± 0.03	88.54 ± 0.15
$\tau_{\text{fixed}} = 0.0300$	98.01 ± 0.04	88.30 ± 0.20
ERL-BCP (adaptive)	$\mathbf{98.36 \pm 0.02}$	$\mathbf{89.70 \pm 0.07}$

where $\tau_{\text{base}} = 0.03$. The measured mean effective timescale across the adaptive ERL trajectories is $\bar{\tau}_{\text{ERL}} = 0.0271$.

Accordingly, the fixed-timescale controls are evaluated at $\tau_{\text{fixed}} \in \{0.0075, 0.0150, 0.0225, 0.0271, 0.0300\}$. The first three values correspond to $0.25\tau_{\text{base}}$, $0.50\tau_{\text{base}}$, and $0.75\tau_{\text{base}}$, respectively, while 0.0271 corresponds to the measured mean adaptive timescale and 0.0300 corresponds to τ_{base}. All configurations use the same 300×1 architecture, balance-derived plasticity rule, bounded inhibitory feedback, optimization settings, and five seed-aligned runs used in Section 4.3. Only the fixed excitatory membrane timescale is varied.

Table C.7 shows a consistent dependence of predictive performance on the selected fixed membrane timescale. Accuracy increases as τ_{fixed} increases from 0.0075 to 0.0271, followed by a slight decrease at $\tau_{\text{fixed}} = 0.0300$. Among the evaluated constant settings, $\tau_{\text{fixed}} = 0.0271$ gives the highest accuracy on both datasets, reaching 98.05% on MNIST and 88.54% on F-MNIST. This setting also corresponds to the measured mean effective timescale of the adaptive ERL trajectories, providing a particularly strong fixed-timescale reference.

Among the tested constants, $\tau_{\text{fixed}} = 0.0271$, which matches the measured mean adaptive timescale, provides the strongest fixed-timescale baseline. ERL-BCP nevertheless improves accuracy by 0.31 and 1.16 percentage points on MNIST and F-MNIST, respectively. The advantage of adaptive regulation therefore persists even when the fixed reference is selected near the empirical mean operating timescale.

Appendix D. Detailed formulation of the dynamical timescale probes

This appendix provides the analytical details of the dynamical probes reported in Section 4.2. The probes are constructed to separately characterize the mapping from current load to the effective liquid timescale and the effect of the timescale on membrane-state evolution.

The probes use $\tau_{\text{base}} = 1$, $\mu = 0.9$, and $D_0 = 0.28$. For the increasing-current-load probe, the common-mode current q increases from 0.04 to 0.96 in increments of 0.092, while the decreasing-current-load probe uses the same values in reverse order. The current load is updated every three displayed steps over $\text{Step} = 0, \ldots, 30$. The elapsed time between successive updates is $\Delta t = 0.42\tau_{\text{base}}$, and q is held constant within each interval. Each probe is initialized at the equilibrium corresponding to its initial load.

Within each interval, the current load is held constant. The membrane state is therefore obtained from the exact piecewise-constant solution

$$u_n = e^{-\Delta t/\tau_n} u_{n-1} + \left(1 - e^{-\Delta t/\tau_n}\right) u_{\star,n}. \tag{D.1}$$

This equation gives the exact state update within each interval of constant current load. It shows that the membrane state u_n relaxes toward the interval-specific equilibrium $u_{\star,n}$ with a finite lag determined by τ_n. A smaller τ_n increases the weighting of the new equilibrium within the same interval, producing faster tracking as the regulated target changes across successive load conditions.

The temporal effect of τ is characterized separately using frozen timescales

$$\tau_j = \tau_{\text{base}} - j\Delta\tau, \qquad j = 0, 1, 2, 3, \tag{D.2}$$

where $\Delta\tau = 0.2\tau_{\text{base}}$. Thus,

$$\frac{\tau_j}{\tau_{\text{base}}} \in \{1.0, 0.8, 0.6, 0.4\}. \tag{D.3}$$

For the step-on response, the normalized trajectory is

$$y_{\text{on}}(t) = \frac{u(t) - u(0)}{u_\star - u(0)} = 1 - e^{-t/\tau}. \tag{D.4}$$

The time required to reach a response fraction p is therefore

$$t_p = -\tau \ln(1 - p). \tag{D.5}$$

Hence, a smaller τ reaches the same normalized response fraction earlier.

For the perturbation-release probe, the state is initialized as

$$u(0) = u_\star + \delta u(0), \tag{D.6}$$

after which the perturbation is released while the background drive and timescale remain fixed. The normalized residual follows

$$y_{\text{rel}}(t) = \frac{\delta u(t)}{\delta u(0)} = e^{-t/\tau}. \tag{D.7}$$

Here, $\delta u(t) = u(t) - u_\star$ is the deviation from the fixed equilibrium, and $y_{\text{rel}}(t)$ gives the fraction of the initial deviation remaining at time t. Once the perturbation is removed at $t = 0$, this deviation decays exponentially toward zero. A smaller τ therefore yields faster recovery toward equilibrium. These analytical relations provide the formulation underlying the dynamical responses presented in Section 4.2.

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